

Chapter 9

A Data Mining Algorithm for Accessing Research Literature in Electronic Databases: Boolean Operators

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ABSTRACT

Searching and retrieving relevant research materials from electronic databases are difficult for many students and early career researchers. Many researchers have abandoned beneficial research projects because they believe that related literature is unavailable to ground their work. This chapter serves as a guide to students, professionals, and internet users on how to pull information from electronic databases easily. The chapter begins by clarifying the concept of electronic databases, the evolution of electronic databases, and the processes involved in indexing scholarly works in an electronic database. The advantages and disadvantages associated with the use of electronic databases are also discussed. The chapter describes how electronic database search works, with insights into some poor practices. The concept of Boolean operators and how they can be used to easily mine desired contents from electronic databases are discussed. The knowledge and use of Boolean operators might become unavoidable in enabling researchers to locate relevant materials for their projects.

INTRODUCTION

The research process includes problem ideation, conceptualisation, literature review, study design, data collection and analysis, and ultimately, disseminating results (in different formats) to publishing outlets. In this chapter, two of these processes are important – literature review and result dissemination. The literature review process entails searching for previously published content from different databases (traditional or electronic). Literature review presupposes that the material to be reviewed is already

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published or disseminated to the public domain. Therefore, a symbiotic relationship exists between literature review and research dissemination. This is because the review of research literature depends on previously disseminated works. Besides, a review of existing literature is required to offer future research directions and sustain dissemination practices.

The published research literature is often managed using physical or electronic databases, with each requiring different methods of assessing indexed contents. To locate research literature from traditional/physical databases, one must visit the library. After that, a catalogue is used to search for such information depending on the form of library classification. In other cases, individuals must go through different files and folders, searching for such information. However, information search and retrieval patterns work differently from the traditional system in electronic databases. This chapter is not aimed at comparing these systems but to discuss electronic databases, their evolution, advantages and disadvantages, the indexing processes and how indexed information can be assessed using a state-of-the-art approach. The emphasis of this chapter is on the Boolean operators due to their growing importance and utility in mining records from large databases or data warehouses. These operators allow for reducing or expanding the number of records returned. Boolean operators may save time by restricting the search scope and eliminating irrelevant results to users' needs.

Therefore, after reading this chapter, readers are expected to be able to: explain the meaning of electronic databases; discuss the evolution of electronic databases; define the term data mining; mention at least four advantages and disadvantages of electronic databases; describe electronic database search works; discuss the indexing processes of scholarly materials in an electronic database; make practical use of Boolean operators in retrieving literature materials from electronic databases.

ELECTRONIC DATABASES**Meaning**

There is no universally accepted meaning for electronic databases due to variations in usage across disciplines, authors, time, and place. However, an electronic database can be defined as an organised collection of information that has been saved on one or more digital media. These materials can be textual or non-textual but are in digital forms to aid easy retrieval and consumption. Stored materials in electronic databases are accessible to users with authorisation and privileges. Electronic databases are used for storing, retrieving, and managing information on a broad range of subjects. Textual materials include books, journal articles, theses, dissertations, patents, conference proceedings, monographs, and other resources stored in digital forms. These resources can be accessed using electronic devices such as computers, smartphones and tablets. On this note, Koedinger et al. (2008) state that e-learning tools, instructional technology, the Internet, and institutional archives of student records have resulted in large data warehouses that educators may access. Therefore, electronic databases are treasures of knowledge discovery that can be investigated and used to help us understand how students learn (Mostow & Beck, 2006). It has also been stated that a large amount of data is available in electronic educational databases that may be used to make better management choices (Bala & Ojha, 2012).

Because of the drive toward open science or open access publications, electronic databases offer a rich variety of scholarly content beyond the offerings of traditional libraries. Besides, in contemporary times, big data has occasioned diverse forms of non-textual materials that can meaningfully be stored

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and accessed electronically. These may take the form of digitally stored audio, visual and audio-visual resources. Since storing and retrieving data from an electronic database is possible, a file system can house small archives on a personal computer. Still, a network of computers or cloud services is required for big repositories. Data structures, efficient information warehousing, programming languages, security/privacy of confidential documents, and networked computing challenges (such as concurrent access and power quality) are all aspects of database architecture. Examples of popular electronic research databases include Directory of Open Access Journals (DOAJ), Educational Resources Information Centre (ERIC), Google Scholar, IEEE Explore, Jstor, Phil Archive, ProQuest, PubMed, ResearchGate, ScienceDirect, Scopus, Web of Science, Mendeley and Zenodo.

Evolution of Electronic Databases

Before computers and databases were developed, every record had to be kept on paper in ancient times. Many lists, notebooks, ledgers, and archives full of tens of thousands, if not millions, of documents, are stashed away in file cabinets (Polding, 2018). Accessing these documents was a time-consuming process that required physical effort. Many issues arose, ranging from lost documents to fires that decimated whole archives, wiping away centuries of history for individuals, groups, and nations. Security issues arose because of the ease with which physical access might be gained. As technology and understanding advanced, whole book communities were consolidated into the first actual “database” (libraries). The primary goal of the libraries was to guarantee that records or information could be stored and accessed conveniently. There is a long-standing tradition in the library world of librarians serving in an intermediary role, whether it be as reference librarians (who help patrons find and use specific resources) or as the people who design the databases and other resources that make it possible for patrons to do just that (Calhoun, 2007). However, in today’s increasingly linked Internet world, information searchers are more self-sufficient, using basic but powerful search engines like Google and expanding beyond library holdings in their hunt for information. This trend led to the surge in the development of different electronic databases hosting a diverse wealth of resources as a traditional library. Each electronic database contains a collection of files and records (UKEssays, 2018) that are carefully collected and curated to meet the needs of several users. The era of big data, characterised by a high volume, variety, value, veracity and velocity of information, ushered in the need for data mining strategies to locate the right information from electronic databases. Boolean operators are an important innovation that enables users to search for information from these large databases across the globe.

Advantages of Electronic Databases Over Manual Databases

- They promote inclusive access to hosted materials.
- Data entry and modification can be done with relative ease.
- They allow for automated data updating and recalculation.
- They are easily accessible compared with traditional archives, eliminating the cost of travelling long distances to physical libraries (Owan et al., 2021).
- Electronic databases can be shared with other software packages.
- Many people may use electronic databases through a network to eliminate duplication of information.

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- Electronic databases provide many pre-programmed capabilities that make calculating data faster and easier.
- An electronic database can be easily backed up and recovered in events of theft, system failure, or system replacement.
- Large amounts of data can be stored in electronic databases.
- Querying, searching, filtering, and retrieving data from electronic databases is simpler than manual databases.
- Electronic databases can be easily protected using passwords for secure and authorised access.

Disadvantages of Electronic Databases

- All users and programmes are dependent on the availability of the database management software. Therefore, the failure of a single component might shut down the whole system.
- Setting up an electronic database is very expensive, depending on the kind of gear, software, and necessary functionality.
- To get the most electronic database, users must possess adequate knowledge. An organisation's success might be jeopardised by a lack of understanding of the system (Mukalele, 2019).'
- Switching to new software and hardware resources and training employees to utilise these new systems is expensive, as is the possibility of hiring specialised people to assist with conversion and system operation.
- Most database management systems are designed to cover a broader scope, not just for a single application. Consequently, certain applications or processes may not load or operate speedily.'
- Sometimes issues of Internet connectivity can deny you access to electronic database resources.
- Some electronic databases are not freely accessible since personal or institutional subscription access is required. Examples include Scopus, Web of Science core collection and subscription-based journals.
- Expensive devices such as computers and other mobile gadgets are required to access electronic databases, limiting inclusiveness to those who can afford them.
- Mobile bandwidth and internet subscriptions from Internet Service Providers (ISPs) are not free, making it difficult for some people to use electronically hosted materials consistently.

Indexing of Research Literature in Electronic Databases

The process of constructing indices for record sets is known as indexing. Metadata about published scholarly materials are stored in an electronic database through indexing. The aim is to enable accessors to identify records or files of interest quickly. Individuals may have to sift through hundreds or thousands of entries without proper indexing to find a single document. Indexing also refers to a collection of phrases, definitions, subjects, and other items ordered in alphabetical order to bring readers to the information they need conveniently. It aids the structuring of material so that readers can quickly identify documents of interest. In practice, scholarly work(s) publication begins with submitting a fully written research report. The submitted work(s) undergoes initial editorial assessment and perhaps peer review. Upon acceptance, the work is published and made public. Published works are stored in electronic databases to promote visibility and accessibility to readers. The publishers may own these databases, or they may be hosted externally hosted for widespread coverage and long-term sustenance.

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However, the indexing of research literature begins after publishing a scholarly work. Nevertheless, other electronic databases (preprint servers) can index academic work before formal publication. Preprint databases promote the timely availability and use of research materials before the peer review process (Chung, 2020; da Silva, 2017; Okon & Ubi, 2021) or while they undergo peer review (Bassey & Owan, 2019). The indexer typically gets a batch of publication page prototypes for post-publication indexing when the work is undergoing final proofing. The indexer will need page duplicates to create a list of headings and subheadings (terms to be included in the index), as well as the position of each essential source. After the preliminary index is completed, it is adjusted for structure, legibility and coherence; formatted according to requirements, proofread and sent to the scholar/researcher as a final soft copy. Indexing takes time and is dependent on the length of the document. The more material a text has, the longer it will take to index. Occasionally, drawings, diagrams, photographs, or tables are often included during the indexing process. One research material can be hosted in multiple databases so long as there are no copyright infringement concerns associated with doing so.

How Searching Works in Electronic Databases

Once research materials are hosted in electronic databases and granted public accessibility, readers' retrieval of indexed information is another concern. In a typical analogue library, interested readers physically visit libraries. Documents are sourced manually using catalogues and library classifications with the help of metadata (such as subject heading, call number, author, and other publication properties). Searching for materials under the traditional system is Herculean and time-consuming. Besides, in cases where some materials are unavailable due to lending, interested users may have to wait until they are returned. On the contrary, the search for published materials in electronic databases is relatively seamless but requires the application of some strategies to get the desired results. In searching for electronically hosted materials, keywords and phrases are used, with shorter terms providing a larger pool of results. The reader must go through each search result to identify materials that match the information of interest. Several users can simultaneously access and utilise one file or document in electronic databases. However, experience has shown that most people do not understand how searching for materials in electronic databases works. As a result, they enter a long strand of phrases or sentences, hoping to get literature materials that exactly match their search.

Let us consider a hypothetical scenario where a doctoral student is working on a dissertation topic - *COVID-19 and Students' Use of Digital Platforms for Online Learning in Nigerian Public Universities*. Most students will enter the full dissertation topic as it is to an electronic database and expect to get an exact result. Unfortunately, the emerging results will be fewer and not likely to be precise. Most of the results will be based on different combinations or permutations of phrases or keywords from the topic. These might include works on *Covid-19 and Students*, *Covid-19 and digital platforms*, *Covid-19 and online learning*, etc. A query in an information retrieval system does not identify a unique item in a collection. Rather, numerous documents usually tally with search queries but at varying degrees of significance (Lashkari et al., 2009). Information retrieval systems typically assign a numerical score to each database item depending on how well it matches the search query. Afterwards, the user is presented with the top-ranked items. If users want to fine-tune their query, they may repeat the procedure or filter the results based on other attributes. This chapter introduces readers, especially early career researchers or emerging scholars, to the art of literature review using a data mining algorithm for more specific and meaningful results from electronic databases.

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THE CONCEPT OF DATA MINING

Data mining is the act of pulling hidden information, unforeseen arrangement, and innovative guidelines from massive datasets or databases. It is a fundamental component of data warehousing and database knowledge discovery. According to Romero et al. (2010), data mining is focused on developing, analysing, and applying automated techniques for detecting patterns in big sets of educational data that would be exceedingly difficult to evaluate in person due to the massive number of records included. Because of the important role that data mining (from huge databases) plays, many scholars have identified it as a key part of emerging fields such as artificial intelligence, machine learning, database management, and data science. The use of data mining is also useful to many data-driven businesses (Ming-Syan et al., 1996; Romero & Ventura, 2012). As a result, biominer is gaining much attention in library and information science (Cheng & Liu, 2019; Liu, 2018; Fernandez-Morales & Bonilla-Carrión, 2020; Nicholson, 2006).

When data mining approaches are used to pull information from electronic databases in research and library science, it is generically referred to as ‘text data mining.’ This name is given because most published scholarly materials are usually in textual form, with some graphical contents usually embedded in them. However, textual data or materials retrieval is embedded in all associated graphic contents (like images, tables, diagrams). In some cases where audio, video, presentation or media files are associated with textual data, text data mining could be used to get secondary links to these media files. Text mining may be pursued when data mining techniques and methodologies extract knowledge from textual data formats. Several methods may provide important information individually or combined with other text data mining approaches. Text data mining has seen several successful applications in a variety of fields. However, several deep sentiment text analysis jobs remain unsolved (Yassir et al., 2020). The use of text mining or “text analytics” can enable users to extract useful information from textual materials. It is common to use text mining to categorise and cluster words and phrases, extract concepts and entities; provide comprehensive classifications; analyse sentiment; summarise publications; and model the mappings. This chapter describes a widely adopted approach to literature search – the Boolean operators. Although the Boolean operators are popular in review studies, only a few scholars have conceived of their application as a text data mining activity (Aboalsamh, 2008; Aoga et al., 2019; Arya, 2021; Bollmann-Sdorra et al., 2003, Ketenci & Gol, 2019; Lv, 2020; Mohesh, 2019; Nguyen, 2005; Pazhaniraja et al., 2020).

THE BOOLEAN OPERATORS

Boolean operators are one of the first data retrieval methods since it uses precise matching to locate documents that satisfy a user-inputted “query.” The queries are matched against documents that contain the search terms. The adjectival term “Boolean” relates to the application of Boolean algebra, which entails the logical combination of words using the Boolean operations *AND*, *OR*, and *NOT* (Lashkari et al., 2009). Boolean *AND*, for example, signifies that both propositions *x* and *y* must be true. Since both claims are in Boolean *OR*, at least one of them must be true. Three Boolean operators may combine any number of logical assertions into one. On the contrary, the Boolean Model is a basic information retrieval method that utilises the set theory. In terms of which words are more essential than others, there is no evidence (weights are binary, either 0 or 1). Using set theory language in mathematics or probability, *AND* operator *intersects* two sets of words or phrases ($A \cap B$), *OR* unites two sets of words or phrases ($A \cup B$); while *NOT* is an inverse form of a set or the difference between two sets.

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Search engines, databases, and online catalogues may benefit from Boolean instructions. Boolean operators *AND*, *OR*, and *NOT* are the most often used in data mining from electronic databases. Parentheses, truncation, and phrases are some of the other available operations. The use of Boolean Operators for the inclusion, combination or exclusion of keywords or phrases in a search yields more systematic and accurate results. This reduces the time and effort spent searching for literature by minimising the number of entries that must be reviewed before being dismissed. Using these operations may minimise or increase the number of records returned. Boolean operators can promote time-saving by restricting queries to include only relevant results while removing irrelevant ones. Some databases and search engines uniquely use Boolean operators, whereas others need upper case letters or punctuation. Some databases (e.g., Google) also automatically insert Boolean operators for users where they are not used. This is done by attempting to guess the user's intention. However, for effective and optimal results, users should give appropriate commands of their intentions to the database rather than rely on guessed or random assignments of operators (that are often invisible). The automatic insertion of Boolean operators may yield undesired results since the database does not understand people's intentions. Sadly, this happens in natural language search, where many users enter their search phrases casually while searching for materials.

Types of Boolean Operators

AND Operator

The *AND* operator demands that all words appear in each item returned. The item is excluded from the resultant list if one word appears in the text but the other does not. The *AND* operator narrows the search scope because the combination in the search string must all be present in the document. For example, a search on *Research AND Productivity* will provide results including both words. Any document that mentions *Research* but has no mention of *Productivity* is not included, and vice versa. The *AND* operator can handle many search queries or pairs beyond just two words. For example, a search might look like this *Mentorship AND "Research Productivity" AND Nigeria*. This search string will only bring out documents that contain the three phrases or terms in no order of permutation. This can be used by a researcher interested in getting literature materials on mentorship and research productivity but focusing on Nigerian studies or authors. If a document contains only two of these terms (for instance, "*Research Productivity*" and *Nigeria* but no *Mentorship*), it is not included in the search results.

OR Operator

The *OR* operator restricts search results to include all the keywords presented in the search string or phrase entered by the user. It expands the scope of search results, unlike the *AND* operator. The *OR* operator is used in cases where a targeted keyword has other synonyms that are of interest to the user and which other scholars may have used. For instance, if the target phrase is the word "*teacher*", a researcher may be interested in including other related terms such as "*instructor*", "*foster parent*", "*schoolteacher*", "*tutor*", "*pedagogue*", and so on to the search results. These terms must be preceded with the word *OR* (preferably in capital letters) to use the *OR* operator. For example, the search string *Teacher OR Mentor* will yield results of all documents in the database containing either "*teacher*" or "*mentor*" or both.

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NOT

The first phrase is searched when this operator is used, but records including the term placed after the operator are removed from the results. A search that attempts to be more specific but excludes relevant results might be harmful. There are several ways to make searches using the *NOT* operator, including double quotes (“ ”). For example, a search string “*Library Science*” *NOT* “*Library Studies*” will include all documents containing the phrase *Library Science* but exclude those containing the phrase *Library Studies*. Just as the *OR* operator is used to capture related terms, the *NOT* operator may be used to refine search results by excluding unwanted associated terms. The *AND NOT* operation can perform the same function as the *NOT* operator.

Double Quotation Marks (“ ”)

This operator is used to enclose a word or phrase so that exact matches are returned in the search results. In a normal search situation (natural language search), typing a group of words (phrases or sentences) may bring several unwanted results for each term in the expressions. This usually leads to search results containing irrelevant or unwanted documents. However, when quotation marks are wrapped around a group of words, the search result must be confined to produce documents containing the phrase in the same order as they appear in the search string. For example, searching for the phrase *Service Delivery* (without quotation marks) can result in either one of *Service*, *Delivery* or *Service Delivery*. The research results might be unnecessary to a researcher interested in literature materials with the mention of the phrase *Service Delivery* and not the other two. The phrase must be placed within the quotation mark to restrict the search to only documents containing the exact expression. That is, “*Service Delivery*.”

Parenthesis ()

Search results will be more relevant to the subject if parenthesis is used to encapsulate search strings. Search engines typically implement the parentheses-enclosed search phrases and then execute the rest of the sentences. Example: A search on (“*Data Mining*” *OR* “*Data Warehousing*”) *AND* “*Information Retrieval*” returns articles containing *Data mining* and *information retrieval*; *Data warehousing* and *information retrieval*; but does not produce documents on *Data mining* or *Data warehousing* when *information retrieval* is not mentioned.

Asterisk (*)

The asterisk is a truncating Boolean operator, usually appended after a word. It is mostly used to get different word forms that other writers may have used. This can be used to obtain related or alternative forms of keywords such as plural, nouns, verbs or adjectives of a word. Unlike other Boolean operators mostly used as conjunctions, the asterisk is usually placed after a truncated keyword but can also be placed before a keyword. When placed after the root word, the search result will yield materials containing both the root word and other possible extensions that can complete the truncated part of the keyword. In other words, the asterisk is attached after a stem word to search for documents containing other words that have that stem and beyond. Therefore, the user will receive results with various endings, but the stem remains present. For example, searching for *Fact** will yield results of documents containing the

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word *Fact* and others extending from it (such as *facts*, *factful*, *facticity*, *faction*, *factiously*, *factitious*, *factitive*, *factoid*, *factorable*, *factorial*, *factories*, *factorise*, *factors*, *facts*, *factual* and so on). As you will have noticed, the stem word (*fact*) is contained in each resulting result but with different modifications after the root word. When the asterisk is used before the stem word, the search result is almost the same as putting the search string in double quotation marks.

The Minus Sign (–)

The minus sign is used primarily to obtain accurate search results by eliminating unwanted documents. The elimination is done by filtering out unwanted keywords (noise) from an initial search result, leading to refined results. For instance, an initial search of the string “*Teachers’ effectiveness*” on the Google Scholar database yielded results containing the search string but with other keywords (noise) such as *students*, *students’ academic performance*, etc. Let us assume that we are not interested in documents discussing *teachers’ effectiveness* in the context of *students* or *students’ performance*. Then the use of the minus sign can enable us to eliminate further documents that mention *teachers’ effectiveness* and *students’* or *students’ performance*. To do this, we will enter the search string as (“*Teachers’ effectiveness*” –*students*) or (“*Teachers’ effectiveness*” –*students’ performance*) depending on whether we are eliminating *students* or *students’ performance*. There is one space after the closing quotation before the minus sign, and there is no space after the minus sign.

Using File Extensions

A file extension is a suffix that comes after the title of a document, usually after a full stop (.) sign. It helps us identify a file type and the difference between classes of files. Normally, a full stop (period) separates the extension from the remainder of the filename, although, in other systems, it is separated by spaces. Very popular file extensions include *.mp3* (for music files), *.docx* (newer versions of Microsoft Word documents), *.xlsx* (newer versions of Excel documents), *.pdf* (for portable document files), *.html* (for hypertext mark-up language files), *.mp4* (for video files), *.exe* (for windows executable files), *.pptx* (for PowerPoint files), *.txt* (for text files) and so on. After the search string, the extension of interest is added to use file extensions for mining data from electronic databases. Doing this will restrict the search only to bring out documents containing the keywords of interest but of a particular file format. For example, if you are interested in getting only PDF files that mention the phrase *Research Metrics*, you should enter the search string “*Research metrics*” *.pdf* into the database. The search result will be all PDF documents containing the phrase *Research Metrics*; other document types such as HTML, Text files and so on will be excluded. Therefore, the use of file extensions narrows search results.

Double period or ellipsis (..)

The double period is used to restrict search results of a keyword or search phrase to a range of periods. This can be very useful for researchers writing review articles or individuals seeking to narrow their search to a period. It can also be useful to students working on a thesis with a specified date range of expected references. When a range is specified, search results will omit documents outside this range, resulting in a more refined search. For example, a researcher can use the search phrase “*Teachers’ effectiveness*” 2017.2022 to get documents that talk about *teachers’ effectiveness* published in 2017, 2018,

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2019, 2020, 2021 and 2022. Works published earlier are not included in the search results. Although this may not work in all databases, it works well in the Google search engine. However, other databases such as Scopus, WoS, and Google Scholar, have handy filters that can enable you to restrict your search results.

Title Search

The title search allows users to search for keywords primarily found in the title of a document. This approach is useful for refining search queries to provide the desired keyword or phrase results only when included in the title. To use the title search, the word “*TITLE*” (without quotes and preferably in capital letters) must be entered first, followed by a parenthesis containing the keyword or phrase of interest. The result will include only documents carrying that keyword or search string in the title. For example, searching for *TITLE (Self-esteem)* will provide results of all documents containing the word *self-esteem* in the title. Unlike other operators that can yield results, if inputted keywords are found in the document’s body, title search restricts search results to only document titles. From experience, the description of a document or record transcends its title to other areas, such as abstract and author keywords. However, these important sections are ignored when the title search is used. Besides, writers will sometimes employ a catchy title to attract their readers’ attention. Sadly, enticing titles are not always sufficient to communicate the main text’s focus. Therefore, users should always be sure of what they are looking for before using the title search. This does not imply that a title search is unimportant; it only increases the chances of missing important records, although it can provide more accurate or direct results of a user’s search. The ideal application of a title search is to identify known records in an electronic database.

Author Search

The author search is an important way of checking out an author to keep track of what they are publishing or their recent publications. Maybe you met a friend or colleague with the same research interest at a conference. If you want to know about their publication history, perhaps for a potential collaboration, the author search can enable you to achieve that. The author search is used to locate documents or records about an author of interest. The author search can be used primarily in two ways: (1) to locate all the documents by an author in a database and (2) to locate specific documents of an author in a database. To locate the documents of an author in a database, a user needs to specify the command, followed by the name of the full name of the author or other possible variants in quotes. This can be performed by entering the command *Author: “Name of the author”*. There is no need for a search keyword or phrase other than the author’s name in question to do this. For example, a user interested in locating all the documents of the author Valentine Joseph Owan could enter the command *Author: “Valentine Owan” OR “V Owan” OR “VJ Owan” OR “Valentine Owan” OR “Valentine Joseph Owan” OR “Owan Valentine Joseph” OR “Owan VJ”*. Note that all the possible name variants of the author have been used with the *OR* operator (previously discussed) to broaden the search into capturing documents where these different name variants may have been used for the same author.

In the second case, if a user is familiar with an article or a document of an author and wishes to locate the exact document, the author’s name must be combined with the work’s title or keywords. In cases where the full title is forgotten, the author’s name with a keyword or search phrase of the document in question could be used. For example, a user is interested in locating an article on *research productivity* written by *Valentine Owan* but cannot remember the work’s full title again. An author search could identify the

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work, including the recalled phrase. In the example above, the user can search for all documents written by *Valentine Owan* containing the phrase *research productivity* by entering the command *Author: "Valentine Owan" AND "research productivity"*. Note that the *AND* operator intersects the author's name with the search phrase, refining the search to the most relevant or related documents. Another way is to enter the command *"research productivity" author: "Valentine Owan"*. This will yield the same results as the previous string. To be more thorough in the search, a user may wish to combine two more author name variants alongside the keyword or phrase. For example, *"research productivity" Author: Author: "Valentine Owan" OR "Valentine Joseph Owan" OR "Owan Valentine Joseph."* Alternatively, the *AND* operator can also be used to achieve the same result such as *"Author: Author: ("Valentine Owan" OR "Valentine Joseph Owan" OR "Owan Valentine Joseph") AND "research productivity"*.

Limitations of Boolean Operators in Accessing Research Literature

- The search is only restricted to documents available in a database. This implies that it cannot provide search results for works not indexed in the database.
- They do not guarantee that search results will always be useful to the researchers. Therefore, the user requires screening and filtering to identify relevant works and ignore unnecessary documents.
- For effective use, sufficient proficiency is required. This implies that individuals must master the art of using these operators for effective leverage.
- No matter how individuals filter their searches using Boolean operators, there will always be noise (unwanted results) in search results. This means that individuals should check through each search result carefully before selection.
- Even though they assist in finding relevant documents through a refined search, it does not indicate which search results are more important than others. Therefore, the user still needs to select useful information through individual search results.
- While they can work very well for some offline electronic databases (Mendeley and Windows Explorer), their use is predominantly Internet-based. Therefore, Internet access is required to connect to online electronic databases.
- In cases where different synonyms are enclosed in parenthesis and separated with the *OR* operator, equal opportunity is not given to these synonyms. Usually, the last keyword or search string in the series of synonyms receives the greatest priority. Therefore, users should remove the last keyword after each assessment and selection round before searching again to optimise search results.

CONCLUSION

This chapter discussed the concept of electronic databases and how to access scholarly literature materials from electronic databases. Special attention was given to using Boolean operators as a data mining algorithm in refining the search results from electronic databases for optimisation purposes. Different Boolean operators were described and how they can be used in accessing scholarly materials. In conclusion, electronic databases are very useful and will be the future of literature review, especially as many traditional libraries are gradually gaining an online presence. Furthermore, the fact that many journals, books, and conference proceedings are slowly taking electronic forms suggests an urgent need for effective strategies to mine these resources from different databases. The knowledge and use of Boolean

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operators might become unavoidable in enabling researchers to locate relevant materials for their projects. However, while Boolean operators can be used individually, a combination of different operators can be implemented for more refined search results from electronic databases. Therefore, researchers, scholars, and scientists should practice the art of using (a combination of) Boolean operators for optimal results. This chapter is useful to all scientists, researchers and scholars, especially those without advanced Internet searching skills.

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KEY TERMS AND DEFINITIONS

Data Mining: Data mining is the process of seeking patterns in large databases and extracting relevant information using machine learning or algorithmic and database techniques.

Electronic Databases: An “Electronic Database” is a computer-maintained collection of data such as peer-reviewed journal papers, dissertation, books, and others, with searchable areas.

Indexing: Indexing is the process of using index words or other symbols to identify, summarise, or otherwise improve the discoverability of a document’s content.

Information Search: This is the process of making conscious efforts to use systematic procedures in going through a large amount of information to locate those that are interesting.

Library: This is a room, series of rooms, or a structure where books may be read or loaned.

Literature: Literature refers to publications in fields such as the natural, behavioural, management, library, and social sciences or humanities that give new empirical or theoretical information for scientific progress or problem resolution.

Metadata: Metadata are records that explain other material, giving a systematic guide that aids in classifying and identifying the characteristics of the material it represents.

Operator: An operator is often a linkage or expression that converts items from one domain into elements from another.

Publications: This refers to the total amount of scholarly works in form of journal articles, theses, book chapters, books among others, that an author/researcher/scientist has contributed to the literature. It also refers to the total number of documents made publicly available by a publishing firm at any given time.

A Data Mining Algorithm for Accessing Research Literature in Electronic Databases

Research: Research refers to any work aimed towards increasing the body of existing knowledge, conducted in a methodical and innovative manner. It is also the process of gathering, organising, and analysing data to better understand a subject or situation.

Search String: A search string is a set of keywords carefully combined with Boolean operators or truncation symbols to improve the chances of individuals locating the desired digital record from an electronic database.